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 Codeforces and Polygon may be unavailable from [August 20, 16:00 \(UTC\)](#) to [August 21, 1:00 \(UTC\)](#) due to technical maintenance.

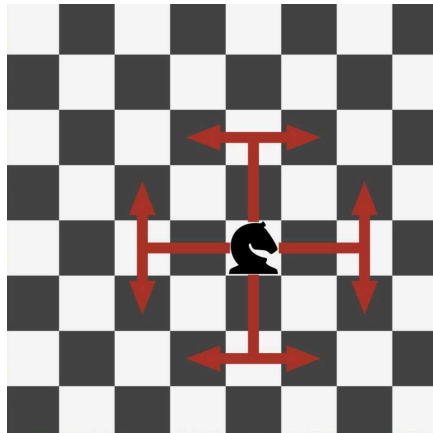
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C. The Knight of Königsberg

time limit per test: 15 seconds

memory limit per test: 512 megabytes

You are given an $N \times N$ chessboard and the initial position (x_1, y_1) of a Knight piece. You need to find the minimum number of moves the Knight needs to reach the target position (x_2, y_2) . If it is not possible to reach the target, print -1 .



Moves of a Knight in Chess

The Knight can move one step in any of the 8 possible directions as shown in the picture.

Input

The first line contains an integer $(1 \leq N \leq 2 \times 10^3)$ — the size of the chessboard.

The second line contains four integers $(1 \leq x_1, y_1, x_2, y_2 \leq N)$ — the initial position (x_1, y_1) and the target position (x_2, y_2) of the Knight on the chessboard.

Output

Print the minimum number of moves the Knight needs to reach the target position. If it's not possible, print -1 .

Examples

input	Copy
3 1 2 1 3	
output	Copy
3	
input	Copy
3 1 1 2 2	
output	Copy
-1	

CSE221 Assignment 06 Summer 2026

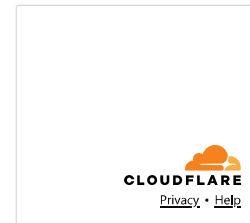
Contest is running

3 days

Contestant



→ Submit?

Language: Java 21 64bit ▼Choose file: Choose File No file chosen

Submit

input	Copy
10 8 4 3 1	
output	Copy
4	

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